Quiz 8: Fibers & Chromatography

Review of Session 8. ~15 minutes.

Section A: Fiber burn test

For each described result, name the class (animal / vegetable / synthetic) and a likely specific fiber:

1. Curls away, burns slowly with sputtering, self-extinguishes, smells like burning hair, black crushable bead.
2. Ignites readily, burns steadily with an afterglow, smells like burning paper, light gray ash, no bead.
3. Shrinks and melts away from the flame, drips, hard shiny bead, sharp chemical smell.
4. Burns very fast with almost no melting, smells like paper, little ash.

Section B: Fiber facts

1. Most synthetic fibers are made from what raw material?
2. Rayon and acetate are made by regenerating what natural material?
3. What kind of fiber is best for moisture-wicking athletic wear, and how does the wicking work?
4. What is the difference between direct and secondary fiber transfer?

Section C: Chromatography

1. Rf = ______ ÷ ______.
2. What is the range of possible Rf values?
3. What is the stationary phase in paper chromatography? The mobile phase?
4. Why must the start line be drawn in pencil, not pen?
5. A dye spot moved 3 cm; the solvent front moved 12 cm. What is the Rf?
6. Who is credited with inventing chromatography, and what does the word mean?
7. To match a crime-scene ink to a suspect's pen, what do you compare?

Answer Key

Section A

1. Animal — wool (or silk).
2. Vegetable / cellulose — cotton or linen.
3. Synthetic — polyester or nylon.
4. Rayon (regenerated cellulose).

Section B

1. Petroleum (and historically coal).
2. Cellulose (from wood pulp).
3. Synthetic (polyester, polypropylene); the channels *between* the fibers pull sweat outward by capillary action, spreading it to the surface to evaporate.
4. Direct = source straight to victim/suspect; secondary = source → person A → person B.

Section C

1. distance the component moved ÷ distance the solvent front moved.
2. 0 to 1.
3. Stationary = the paper; mobile = the solvent.
4. Pen ink would dissolve and run up the paper.
5. 3 ÷ 12 = 0.25.
6. Mikhail Tsvet; "color writing" (Greek *chroma* + *graphein*).
7. The number of spots and their Rf values, run under the same conditions.